LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Factors affecting CL, Theories of Lift production, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back and CL
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Factors affecting CL, Theories of Lift production, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back and CL.
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical concepts via instructor-led discussion; Safety: Maintain classroom discipline; Supervision: Instructor monitors trainee engagement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the wing in flight?
(b) Define the term 'Angle of Attack'.
(c) List the four fundamental forces acting on an aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Bernoulli's Principle and Lift Generation • Newton's Third Law and Downwash • Four Forces: Lift, Weight, Thrust, Drag • Pitching Moment and Center of Pressure • Definition of Coefficient of Lift (CL)	L & Ds	PPT, IB	15 min
(b)	• Effect of Aspect Ratio on CL • Effect of Sweep Back on CL • Impact of Angle of Attack on CL • Types of Aerofoil Sections • Conditions of Performance	L & Ds	PPT, PDF	20 min
(c)	• Relationship between CL and Stall Speed • Summary of Lift Production Theories • Practical implications for Air Agniveer • Review of Pitching Moment effects • Transition to Pressure Distribution	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how increasing the aspect ratio affects the coefficient of lift.
(b) Describe the relationship between the center of pressure and the pitching moment.
(c) How does sweep back influence the lift characteristics of a wing?

12. Home Assignment and Follow Up	Prepare a brief note comparing the lift generation theories of Bernoulli and Newton, and explain how these concepts will be applied in the next lesson regarding pressure distribution over an aerofoil.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Study of pressure distribution over an aerofoil at various angles of attack, determination of center of pressure and aerodynamic center. various forces e.g. lift, drag and moments	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min